

Proxy Exploitation in a Minimal Signaling Ecology

How Confounded Evaluation Creates and Redistributes Exploitation Regimes

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The Question

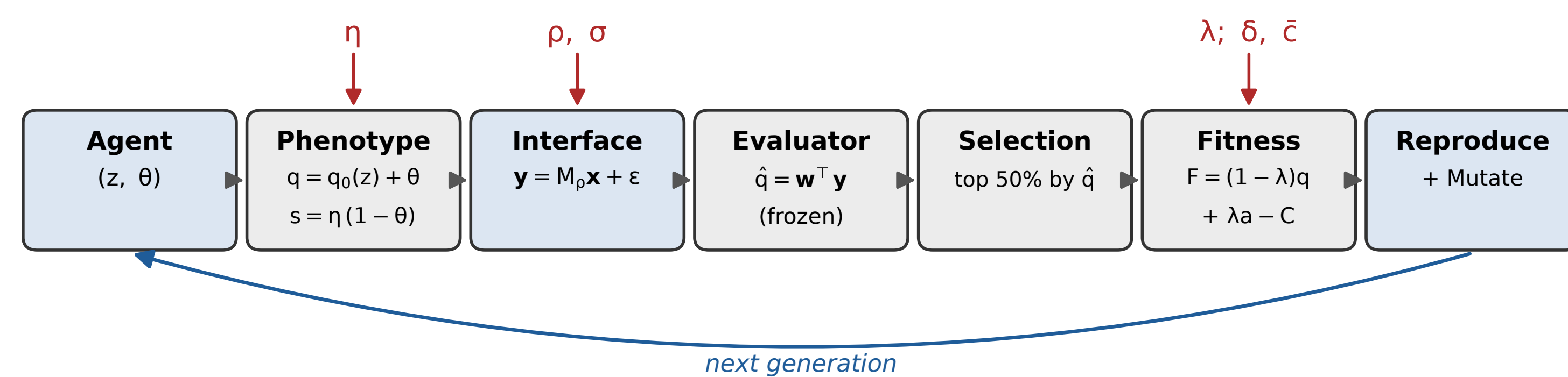
Adaptive agents are selected through an evaluative channel that **confounds genuine quality with surface appearance**.

- How does the system respond?
- Can the failure be *repaired* by keeping the evaluator current?

Four findings

- Under a frozen evaluator, **three distinct regimes** emerge.
- A **two-stage mechanism**: miscalibration initiates failure, demographic takeover sustains it.
- Periodic recalibration does **not remove** the exploitation niche — it **redistributes** failure modes.
- The niche is **distribution-robust** across five prior families.

Model

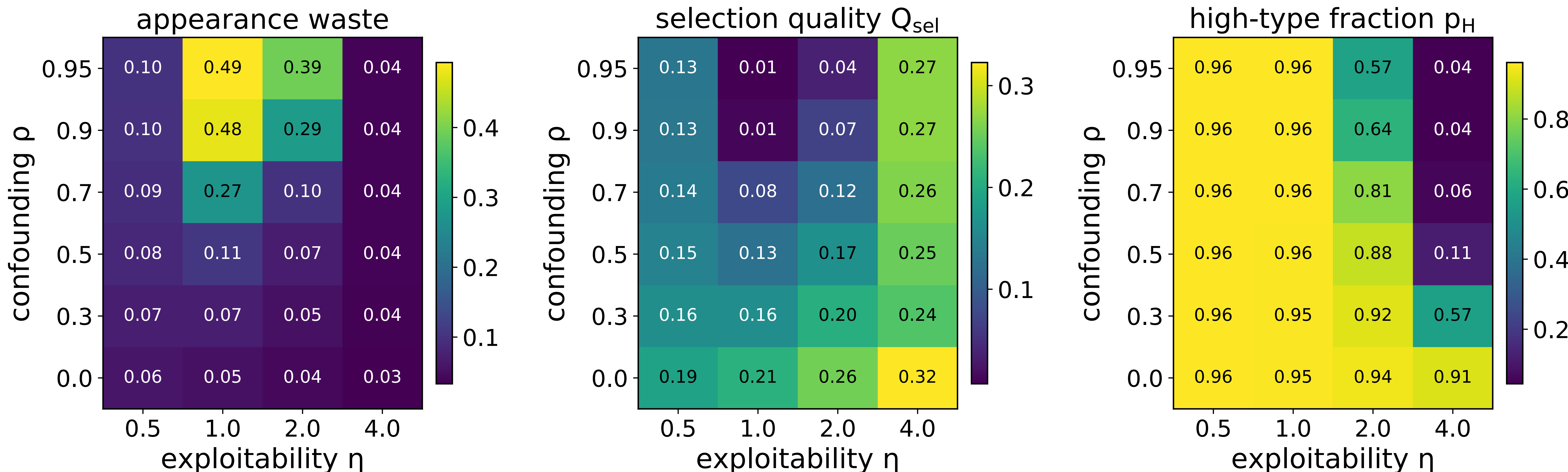


Agents split a unit budget via θ : genuine quality $q = q_0(z) + \theta$ vs. appearance $s = \eta(1 - \theta)$. A single parameter ρ sets channel confoundedness; a *frozen* MMSE evaluator scores $\hat{q} = \mathbf{w}^T \mathbf{y}$ and the top 50% reproduce.

$$\kappa \equiv g_1 - \eta g_2 \quad (\text{proxy-alignment margin})$$

$\kappa \approx 0 \Rightarrow$ quality and appearance are **score-indistinguishable**. ~62,000 runs • $N=200$, $T=2000$ • 200 runs/cell • steady state = last 500 generations

Three Regimes Under a Frozen Evaluator



1. Aligned ($\rho=0$)

waste $\lesssim 6\%$, $Q_{\text{sel}} \gtrsim 0.19$ for all η — no confounded direction to exploit.

Peak exploitation sits at *intermediate* η , not the highest: a bounded window near $\eta=1.25$ – 1.5 .

2. Proxy capture ($\rho=0.95$)

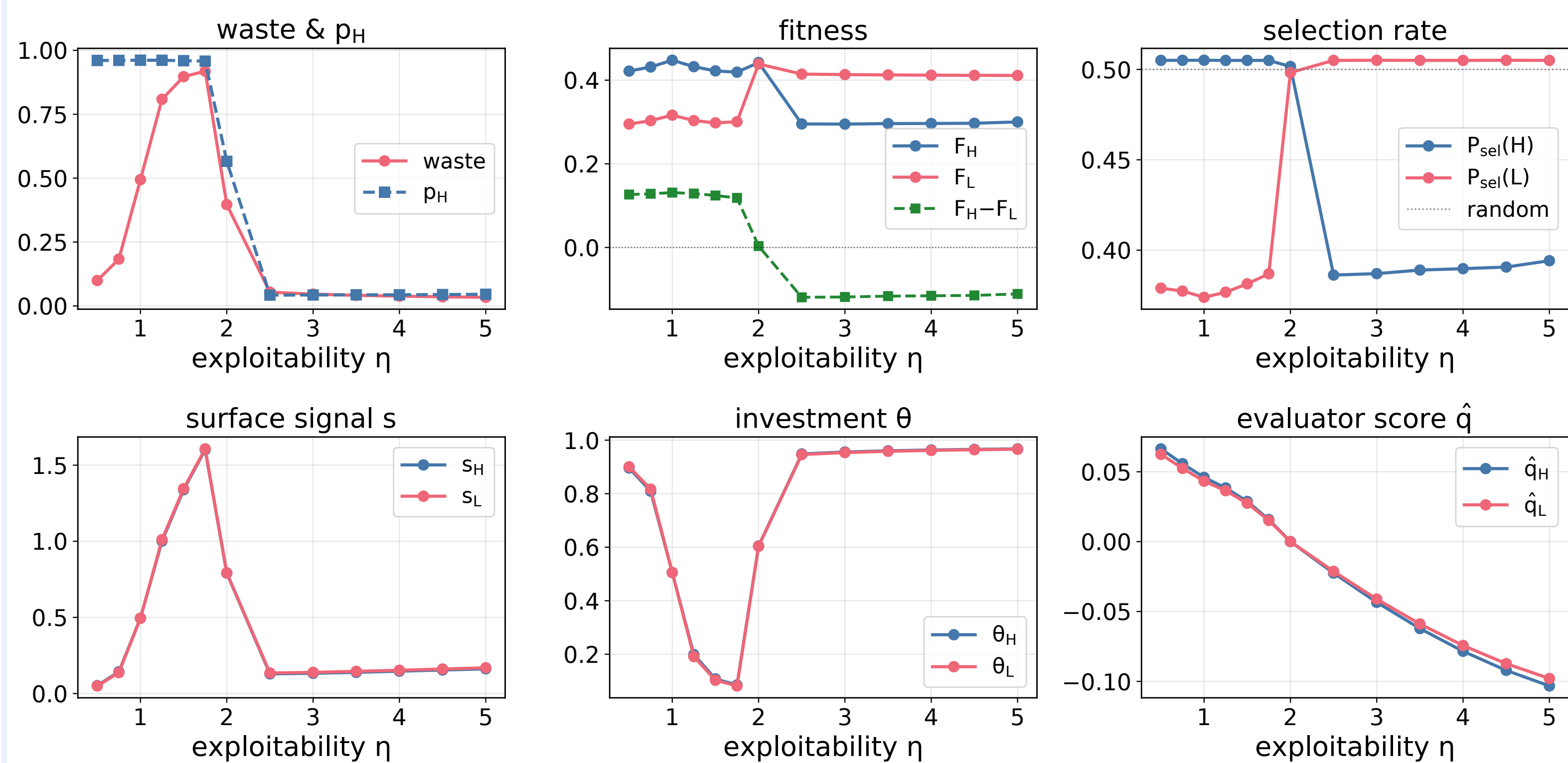
waste 49% at $\eta=1$, peaking **89%** near $\eta=1.5$; p_H holds at 0.96 while Q_{sel} collapses to **0.007**.

3. Low-quality dominance ($\eta=4$)

waste drops, but p_H collapses to **0.04** — deception becomes unnecessary.

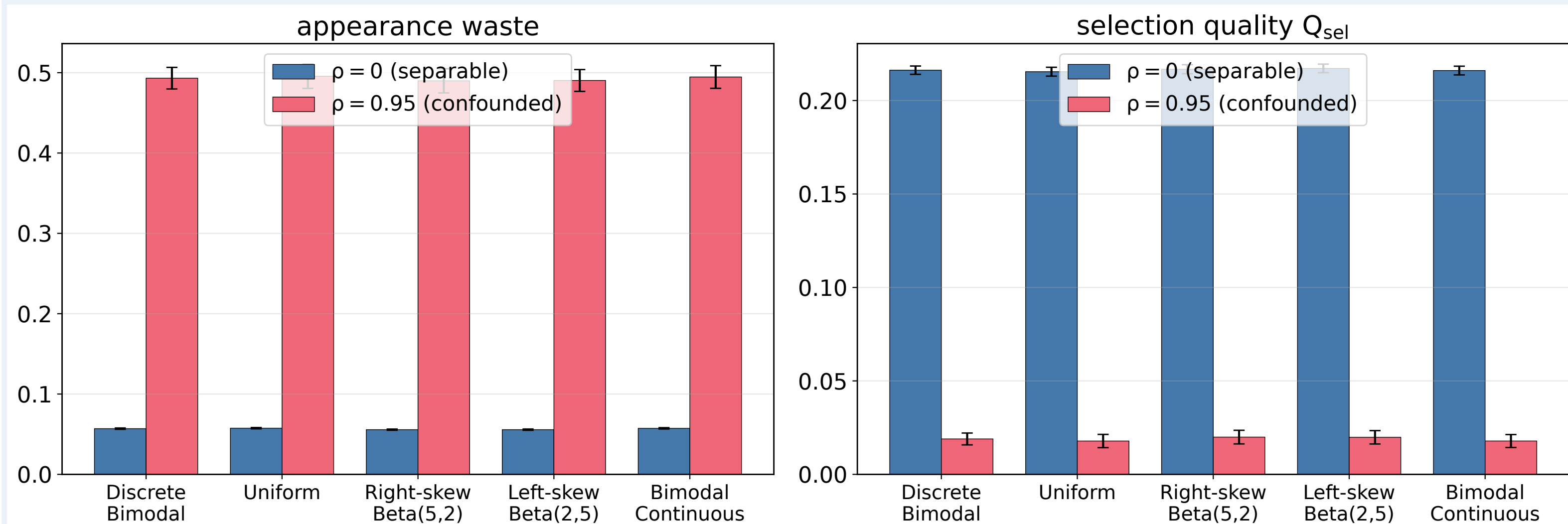
Two-Stage Mechanism ($\rho=0.95$)

η sweep: mechanism dissection ($\rho=0.95$)



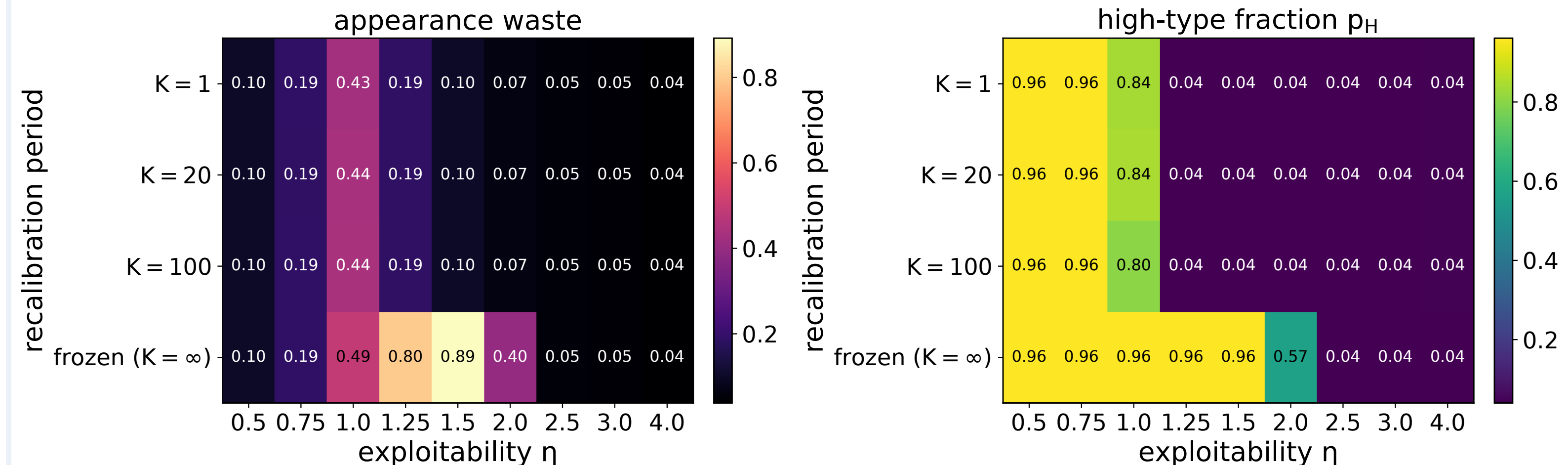
- Both types converge to $\theta \approx 0.96$; the evaluator loses discriminative power (at $\eta=4$: R^2 0.85 $\rightarrow$ 0.11).
- Selection **inverts**: $P_{\text{sel}}(H)=0.39 < 0.50$, gap $F_H - F_L \approx -0.12$.
- θ variance collapses $\sim 95\%$ ($0.040 \rightarrow 0.0019$); transition sharp near $\eta \approx 2.06$.
- Snapshot refit rescues the capture zone ($Q_{\text{sel}} -0.094 \rightarrow +0.049$) but **not** the dominance zone ($P_{\text{sel}}(H)$ 0.338 $\rightarrow$ 0.339).

Robustness



- Five prior families at $\rho=0.95$: waste 49.0–49.6%, $Q_{\text{sel}} +0.018$ – $+0.020$ (near-identical); separable: waste $\approx 5.7\%$, $Q_{\text{sel}} \approx 0.216$.
- A learned MLP evaluator reproduces the phase diagram (mean $|\text{waste}|=0.005$).
- Information-matched control: confounding costs -0.07 in $\hat{q}$ *beyond* information loss.
- Differential cost δ 1 $\rightarrow$ 3: reliability 0.57 $\rightarrow$ 0.97, yet AUROC only 0.50 $\rightarrow$ 0.61 — the gain is demographic purification, not honesty.

Recalibration Redistributes the Niche



Refit the evaluator every $K \in \{\infty, 100, 20, 1\}$ generations:

- At $\eta=1.25$ – 1.5 the frozen model shows extreme capture (waste 0.80/0.89, $p_H \approx 0.96$). **Any** recalibration ($K=100$ suffices) cuts waste to 0.10–0.19 — but p_H crashes to **0.04**.
- Dominance zone ($\eta \geq 2.5$): all schedules agree to within 0.002 — takeover ignores evaluator freshness.
- Confounding-specific: $\blacksquare$ waste 0.002 at $\rho=0$ vs. 0.135 at $\rho=0.7$ — recalibration only matters when the channel is confounded.
- Replicates under a continuous prior $q_0 \sim U[0.55, 0.75]$: at $\eta=1$, $K=1$ waste 0.436 vs. 0.426 discrete; $Q_{\text{sel}} +0.035$ vs. $+0.030$.

A faster-tracking evaluator does not see *less* exploitation — it sees a **different kind**.

Takeaways

- The object is the **niche**, not the failure mode. Interventions change *which* failure appears, not *whether* one does.
- Recalibration trades a *recoverable* failure (costly deception) for an *irrecoverable* one (collapse of the high-quality population).
- Monitoring proxy investment alone misses the more dangerous transition.
- Structural parallel to reward-model overoptimization: “keep the reward model fresh” may only work inside a narrow operating window.
- Raising the stakes λ is nearly neutral when the interface is separable ($\blacksquare \hat{q} = -0.009$) but destructive under confounding ($\blacksquare \hat{q} = -0.45$).
- Channel *geometry* deserves first-class status in signaling models, alongside cost structure.